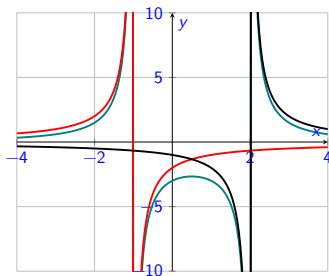


Partial Fractions

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The parts of today's class:

1 News!

- Tutorials
- Assignments

2 Partial Fractions

- 4 cases
- Case 1
- Case 2
- Case 3
- Case 4

3 Towards Limits

4 Exercises

Slides are on canvas, and at
<https://www.niallmadden.ie/>

2627-MA140/



Tutorials start **this** week. See your time-table (or canvas) for the schedule.

Rang teagaisc trí Ghaeilge (Irish tutorial): ...

- ▶ There is currently a “practice” assignment open. See <https://universityofgalway.instructure.com/courses/58477/assignments/162051>
- ▶ During tutorials, the tutor will solve some similar questions. You can access the **tutorial sheet** at the link above (and on Canvas: Modules... Tutorial Sheets).
- ▶ **Problem Set 1** will be ^{open} by Thursday, and will be due by 5pm, Monday 28 September.

.....

Also: try the exercises at the end of each set of lecture slides: they are similar in style and standard to exam questions.

Partial Fractions

Rational Functions have the general form $f(x) = \frac{p(x)}{q(x)}$, where $p(x)$ and $q(x)$ are polynomials.

An (proper) rational function can often be written as a sum of simpler ones: **partial fractions**.

For example

$$\frac{8x - 12}{x^2 - 2x - 3}$$

can be written as

$$\frac{3}{x-3} + \frac{5}{x+1}$$


Check: (*next slide*)

Partial Fractions

$$\frac{3}{x-3} + \frac{5}{x+1} = \left(\begin{array}{l} \text{rewrite so they} \\ \text{have the same} \\ \text{denominator} \end{array} \right)$$

$$\frac{3(x+1)}{(x-3)(x+1)} + \frac{5(x-3)}{(x+1)(x-3)} =$$

$$\frac{3x+3}{x^2-3x+x-3} + \frac{5x-15}{x^2-2x-3} = \frac{8x-12}{x^2-2x-3}$$

Partial Fractions

Note: Any polynomial (with real coefficients) can be factorised fully into the product of

- ▶ linear
- ▶ and irreducible quadratic factors.

Examples:

$$x^2 - 1 = (x + 1)(x - 1)$$

[2 linear terms]

$x^2 + 1$ is irreducible.

We get different combinations of factors in the denominator. Let's look at **four cases**, and how to find the partial fractions in each case.

The four cases

1. Denominator has **linear factors to the power of 1**
2. Denominator has **factors to the power greater than 1 (i.e repeated linear factors)**.
3. Denominator has **irreducible quadratic factors**.
4. Denominator has **irreducible quadratic factors to power greater than 1**.

$$1. \quad \frac{1}{(x-1)(x+1)} = \frac{1}{x^2-1}$$

$$2. \quad \frac{1}{(x-1)^2(x+1)}$$

$$3. \quad \frac{1}{x^2+1}$$

$$4. \quad \frac{1}{(x^2+1)^3}$$

Case 1: Linear factors to the power of 1 in the denominator.

We have **two methods** to find A and B .

1. Comparing coefficients
2. Substituting specific values for x .

Method 1: Comparing coefficients**Example**

$$\frac{3x}{(x-1)(x+2)} = \frac{A}{x-1} + \frac{B}{x+2}. \text{ Find } A, B.$$

$$\text{So } \frac{3x}{(x-1)(x+2)} = \frac{A(x+2)}{(x-1)(x+2)} + \frac{B(x-1)}{(x+2)(x-1)}$$

Now, we need

$$3x = Ax + 2A + Bx - B$$

$$[x^0=1]$$

$$\Rightarrow \left. \begin{array}{l} Ax + Bx = 3x \\ 2A - B = 0 \end{array} \right\} = \begin{array}{l} A + B = 3 \\ 2A - B = 0 \end{array} \Rightarrow A=1, B=2.$$

Method 1: Comparing coefficients**Example**

$$\frac{3x}{(x-1)(x+2)}$$

So this gives that

$$\frac{3x}{(x-1)(x+2)} = \frac{1}{x-1} + \frac{2}{x+2}$$

Method 2: Substituting specific values for x .

Again

$$\frac{3x}{(x-1)(x+2)} = \frac{A(x+2)}{(x-1)(x+2)} + \frac{B(x-1)}{(x-1)(x+2)}.$$

$$\Rightarrow 3x = A(x+2) + B(x-1).$$

Set $x = 1$ so $3(1) = A(1+2) + B(1-1)$
 $\Rightarrow 3 = 3A \Rightarrow A = 1.$

Set $x = -2$ so $3(-2) = A(0) + B(-3)$
 $\Rightarrow -3B = -6 \Rightarrow B = 2$ ✓

Example

Write $\frac{8x-12}{x^2-2x-3}$ as sum of partial fractions.

1. Factorize x^2-2x-3 as $(x-3)(x+1)$.

2. Now find A and B such that

$$\frac{A}{x-3} + \frac{B}{x+1} = \frac{8x-12}{x^2-2x-3}$$

$$\text{So } A(x+1) + B(x-3) = 8x-12.$$

$$3. \text{ Set } x=3 \Rightarrow 4A = 12 \Rightarrow A = 3$$

$$4. \text{ Set } x=-1 \Rightarrow B(-4) = -20 \Rightarrow B = 5.$$

(2) Linear factors to the power greater than 1 in the denominator, (i.e repeated linear factors).

If $(x - \alpha)^k$ appears in the denominator, it will give rise to the following terms:

$$\underbrace{\frac{A_1}{x - \alpha}}_{\text{linear}} + \underbrace{\frac{A_2}{(x - \alpha)^2}}_{\text{quadratic}} + \dots + \frac{A_k}{(x - \alpha)^k}$$

linear quadratic.

In this case we get extra parameters to determine...

Example

Find A , B and C such that

$$\frac{3x+1}{(x-1)^2(x+2)} = \frac{A}{x-1} + \frac{B}{(x-1)^2} + \frac{C}{x+2}$$

(Note: we'll find that $A = 5/9$, $B = 4/3$ and $C = -5/9$).

$$\frac{3x+1}{(x-1)^2(x+2)} = \frac{A(x-1)(x+2)}{(x-1)(x-1)(x+2)} + \frac{B(x+2)}{(x-1)^2(x+2)} + \frac{C(x-1)^2}{(x+2)(x-1)^2}$$

So $3x+1 = A(x-1)(x+2) + B(x+2) + C(x-1)^2$.
Solve for A, B, C .

So $3x+1 = \frac{A(x-1)(x+2) + B(x+2) + C(x-1)^2}{(x-1)^2(x+2)}$.
Solve for A, B, C .

Set $x=1 \Rightarrow$
 $4 = A(1-1)(1+2) + B(1+2) + C(1-1)^2$
 $\Rightarrow B = 4/3$.

Next set $x=-2$
 $-5 = C(-3)^2 \Rightarrow 9C = -5 \Rightarrow C = -5/9$

Then use original eqn to get $A = 5/9$.

(3) Irreducible quadratic factors.

Irreducible quadratic factors can not be factorised using real numbers, e.g. $x^2 + x + 1$.

An irreducible quadratic factor $ax^2 + bx + c$ gives rise to partial fractions of the form

$$\frac{Ax + B}{ax^2 + bx + c}.$$

Example 2.34 from textbook

If one writes

$$\frac{5x}{(x^2 + x + 1)(x - 2)} = \frac{Ax + B}{x^2 + x + 1} + \frac{C}{x - 2}$$

then we find $A = 10/7$, $B = 5/7$ and $C = 10/7$.

(4) Irreducible quadratic factors to power greater than 1.

Each repeated irreducible quadratic factor $(ax^2 + bx + c)^k$ in the denominator will give rise to

$$\frac{A_1x + B_1}{ax^2 + bx + c} + \frac{A_2x + B_2}{(ax^2 + bx + c)^2} + \cdots + \frac{A_kx + B_k}{(ax^2 + bx + c)^k}.$$

These can be done in a similar way to the previous case. But the calculations are pretty messy, so we won't even try!



Towards Limits

When we were considering the domain of a function, we looked at those x -values for which the function was not defined.

Example

$$f(x) = \frac{x^2 - 2}{x - 1}$$

$$g(x) = \frac{x^2 - 1}{x - 1}$$

Neither f nor g are defined at $x = 1$.

But what happens if x gets very close to 1?

Not defined.

x	0.900	0.990	0.999	1	1.001	1.010	1.100
$f(x)$	11.9	101.99	1001.99	Not Def.	-999.9	-99.9	-9.99
$g(x)$	1.9	1.99	1.999	Not Def.	2.001	2.01	2.1

Let's look at the graphs of f and g .

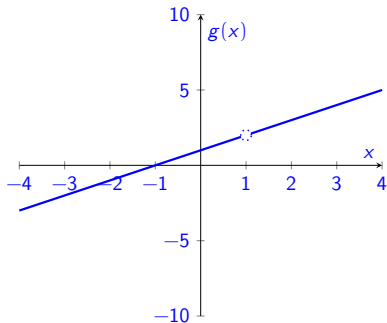
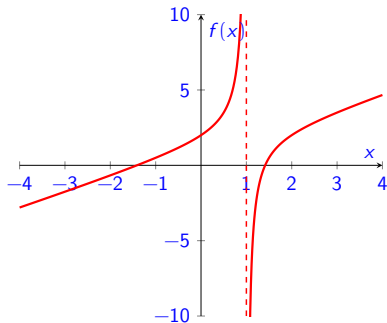
Def.

Towards Limits

Example

$$f(x) = \frac{x^2 - 2}{x - 1}$$

$$g(x) = \frac{x^2 - 1}{x - 1}$$



Finished here Tuesday

Towards Limits

In the previous example, we saw that, although neither f nor g was defined at $x = 1$, they behaved very differently as x approaches 1.

To discuss this we'll need the concept of a **limit** which, roughly, relates to the value of function as it **approaches** a point (but not actually at that point).

$$\lim_{x \rightarrow a} f(x) = L$$

The concept of a limit is a prerequisite for a proper understanding of calculus and numerical methods.

Exercises

Exercise 2.1.1

Find the constants A , B and C , so that

$$\frac{2x + 1}{(x - 2)(x + 1)(x - 3)} = \frac{A}{x - 2} + \frac{B}{x + 1} + \frac{C}{x - 3}$$

Exercises

Exercise 2.1.2

Express the following as partial fractions.

1. $\frac{6}{x^2 - x - 2}$

2. $\frac{2x - 1}{x^2 - x - 2}$

3. $\frac{x - 1}{(x + 1)(x^2 - x - 2)}$

4. $\frac{x}{x^2 + 2x + 1}$

5. $\frac{1}{x^3 - 1}$

Exercises

Exercise 2.1.3 (MA140 Exam, 24/25)

Express $\frac{3x + 1}{x^2 - x - 2}$ as partial fractions.